

**Shri G. S. Institute of Technology and Science Indore**  
**B.Tech. I Year Branch: Mechanical Engineering**  
**IT10007: Fundamentals of Information Technology and Artificial**  
**Lab Assignment - 6**



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| 1.  | Write a program to initialize a one-dimensional integer array and find the largest and smallest element in the array.                                                                                                                                                                                       |
| 2.  | Write a program to count the number of occurrences of a given element (entered by the user) in an array.                                                                                                                                                                                                    |
| 3.  | Write a program to count the number of even and odd numbers in an array.                                                                                                                                                                                                                                    |
| 4.  | Write a program to read two matrices A and B (two-dimensional array of size m x n) and print the following:<br>(a) A + B<br>(b) A – B<br>(c) A x B                                                                                                                                                          |
| 5.  | Write a program to read two strings using fgets function and perform following operations and display the output:<br>1. copy one string to another using strcpy()<br>2. concatenate two strings using strcat()<br>3. compare two strings using strcmp()<br>4. find the length of the strings using strlen() |
| 6.  | Write a program to read two-character string arrays and perform following operations:<br>1. find the length of the strings without using strlen()<br>2. copy one string to another without using strcpy()<br>3. compare two strings without using strcmp()                                                  |
| 7.  | Write a program to calculate the sum of first n odd numbers using function.                                                                                                                                                                                                                                 |
| 8.  | Write a program that defines a function countDigits(int n) which counts and returns the number of digits in a number.<br>Example: Input: 12345      Output: 5                                                                                                                                               |
| 9.  | Write a program using a function to generate Fibonacci series up to n terms. The value of n to be passed to the function.                                                                                                                                                                                   |
| 10. | Write a program using a function to search an element in an array. Pass array and element to be searched (as entered by the user) as parameters to the function.                                                                                                                                            |